Ex. No. 04	UTILIZE THE ASSETS AND MEDIA
Date of Exercise	
Git Hub Host Link	https://github.com/gurubruhh/Flutter-Laboratory.git

Aim

To add and display image, audio and video assets in a Flutter application.

Description

Flutter provides support for adding external media assets such as images, audio files and videos to an application. Assets are stored inside the project and registered in the `pubspec.yaml` file. Images can be displayed using `Image.asset()`, while video files can be played using the `video_player` package. In this experiment, an image and video are added and displayed in the Flutter application with play and pause controls.

Procedure:

1. Create a new Flutter project named `media_app_new`.
2. Create `assets/images` and `assets/video` folders inside the project.
3. Add an image file inside `assets/images`.
4. Add an MP4 video file inside `assets/video`.
5. Register the asset folders in `pubspec.yaml`.
6. Add the `video_player` package using `flutter pub add video_player`.
7. Open `lib/main.dart` and write the program to display the image and video.
8. Use `Image.asset()` to load the image from the assets folder.
9. Use `VideoPlayerController.asset()` to load the video.
10. Initialize the video controller and display the video using `VideoPlayer`.
11. Add a button to play and pause the video.
12. Run the application using Chrome and verify that the image and video are displayed correctly.

Program

```
import 'package:flutter/material.dart';
import 'package:video_player/video_player.dart';

void main() {
  runApp(const MediaApp());
}

class MediaApp extends StatelessWidget {
  const MediaApp({super.key});

  @override
  Widget build(BuildContext context) {
    return const MaterialApp(
```

```
    debugShowCheckedModeBanner: false,
    home: MediaDemo(),
   );
  }
}

class MediaDemo extends StatefulWidget {
  const MediaDemo({super.key});

  @override
  State<MediaDemo> createState() => _MediaDemoState();
}

class _MediaDemoState extends State<MediaDemo> {
  late VideoPlayerController _controller;

  @override
  void initState() {
   super.initState();

   _controller = VideoPlayerController.asset(
    'assets/video/sample.mp4',
   );

   _controller.initialize().then((_) {
    setState(() {});
   });
  }

  @override
  void dispose() {
   _controller.dispose();
   super.dispose();
  }

  @override
  Widget build(BuildContext context) {
   return Scaffold(
    appBar: AppBar(
     title: const Text('Flutter Assets and Media'),
    ),
    body: SingleChildScrollView(
     child: Center(
      child: Padding(
       padding: const EdgeInsets.all(20),
       child: Column(
        children: [
         const Text(
```

```
  'Hello World',
  style: TextStyle(
  fontSize: 28,
    fontWeight: FontWeight.bold,
  ),
),

const SizedBox(height: 30),

const Text(
  'Image Asset',
  style: TextStyle(
  fontSize: 22,
    fontWeight: FontWeight.bold,
  ),
),

const SizedBox(height: 10),

Image.asset(
  'assets/images/image1.png',
  width: 250,
  height: 250,
  fit: BoxFit.contain,
),

const SizedBox(height: 40),

const Text(
  'Video Asset',
  style: TextStyle(
  fontSize: 22,
    fontWeight: FontWeight.bold,
  ),
),

const SizedBox(height: 10),

_controller.value.isInitialized
    ? SizedBox(
        width: 600,
        child: AspectRatio(
          aspectRatio: _controller.value.aspectRatio,
          child: VideoPlayer(_controller),
        ),
      )
    : const CircularProgressIndicator(),
```

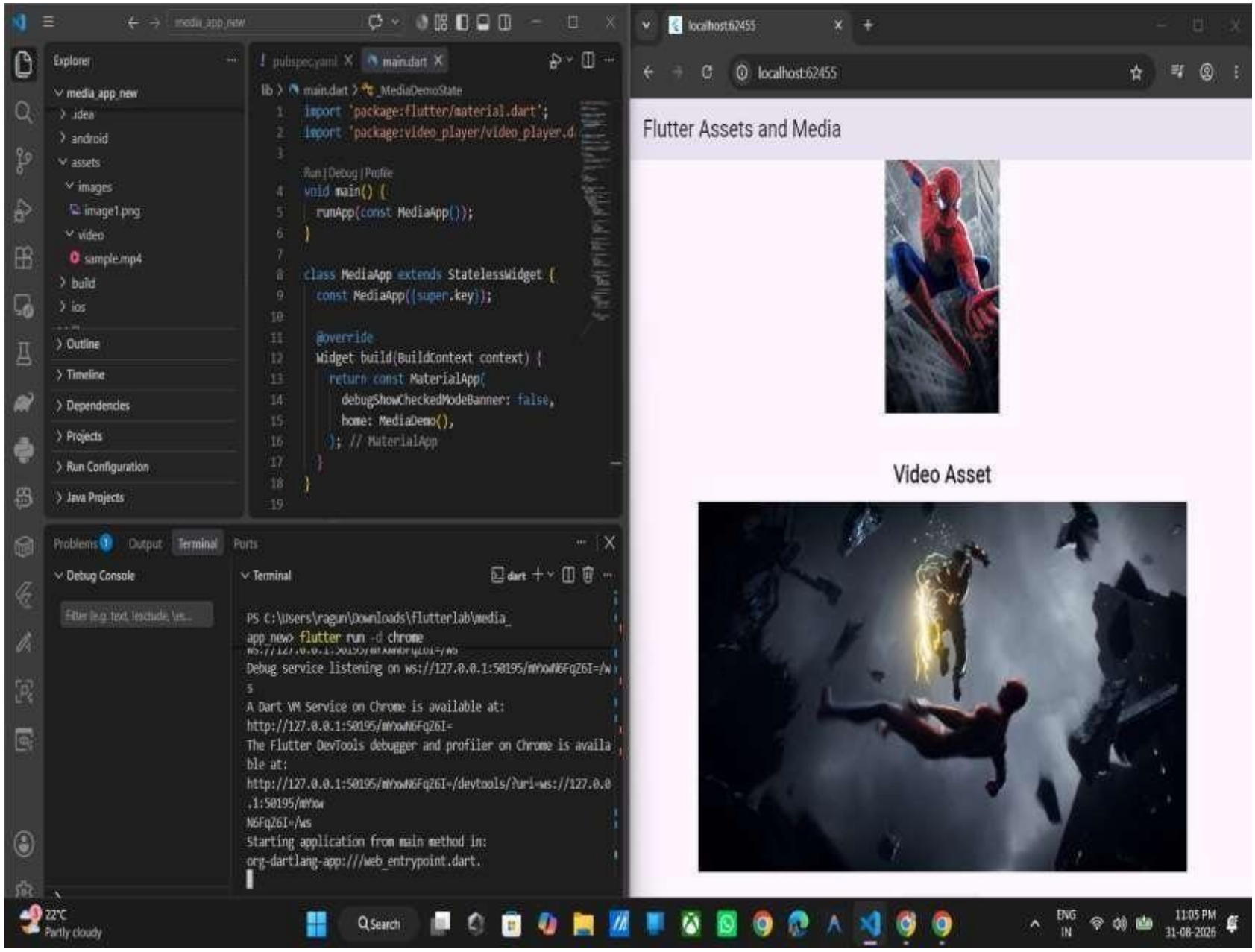
```
            const SizedBox(height: 20),

            ElevatedButton(
              onPressed: () {
                setState(() {
                  if (_controller.value.isPlaying) {
                    _controller.pause();
                  } else {
                    _controller.play();
                  }
                });
              },
              child: Text(
                _controller.value.isPlaying
                    ? 'Pause Video'
                    : 'Play Video',
              ),
            ),
          ],
        ),
      ),
    ),
  ),
 );
 }
}
```

URK23CS1113

24CS2016 Building Applications using Flutter

Output



Result

Thus, the Flutter application was successfully developed to load and display image and video assets. The video was successfully controlled using Play and Pause buttons.